BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DG KHAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: 8467 / 8468

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

OBJECTIVE

Time Allowed: 20 MINUTES | **Maximum Marks:** 17 | **Paper Code:** 8467

Note: You have four choices for each objective type question as A , B , C and D . The choice which you think is correct , fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. Antibody used for treatment of cancer is obtained from

(A) Soyabean
(B) Maiz
(C) Corn
(D) Arabidopsis

Q2. The compound which made environment of earth from reducing to oxidizing is

(A) Carbon dioxide
(B) Nitrogen dioxide
(C) Oxygen
(D) Ozone

Q3. The profession of a species in an ecosystem is called

(A) Habit
(B) Habitat
(C) Niche
(D) Trophic level

Q4. Thar is desert ecosystem of

(A) Punjab
(B) Sindh
(C) Balochistan
(D) Khyber Pakhtoon Khawah

Q5. In sea , tides are generated due to pull of

(A) Earth
(B) Sun
(C) Moon
(D) Supiter

Q6. Large leaves are found in

(A) Xerophytes
(B) Mesophytes
(C) Hydrophytes
(D) Sciophytes

Q7. Opening of buds is due to

(A) Photonasty
(B) Epinasty
(C) Hyponasty
(D) Thermonasty

Q8. The structures help to maintain minerals in the blood

(A) Bone
(B) Muscle
(C) Skin
(D) Gland

Q9. Neurons responsible to carry nerve impulse from central nervous system to effector are

(A) Sensory neuron
(B) Associative neuron
(C) Intermediate neuron
(D) Motor neuron

Q10. Certain human male fail to develop secondary sexual characters due to absence of

(A) Progesteron
(B) Oxytocin
(C) Testosteron
(D) Luteonizing hormone

Q11. Eggs with diploid number of chromosomes are produced as a result of

(A) Normal mitosis
(B) Normal meiosis
(C) Modified mitosis
(D) Modified meiosis

Q12. Apical dominance is caused by

(A) Auxin
(B) Cytokinin
(C) Gibberellin
(D) Ethene

Q13. Complete set of chromosomes in an organism is called

(A) Genome
(B) Genotype
(C) Phenotype
(D) Karyotype

Q14. In a nucleotide , Nitrogen base is attached to carbon number ..... of pentose sugar

(A) 1
(B) 2
(C) 3
(D) 4

Q15. An example of cell that enters G0 – phase permanently during cell cycle is

(A) Gland cell
(B) Skin cell
(C) Nerve cell
(D) Bone cell

Q16. Gene I for blood group is found on chromosome number

(A) 6 (B) 7

(C) 8 (D) 9

Q17. An example of restriction endonuclease is

(A) Taq polymerase (B) ECoR1

(C) Gyrase (D) Ligase

SUBJECTIVE SECTION-I / SECTION-II

Q2. (Marks 16) Write short answers any Eight (8) of the following

i Differentiate between hypotonic and hypertonic environment

ii Briefly write about pyrexia

iii How does high temperature affect plant metabolism ? Write the way plants manage with high temperature.

iv Compare epinasty and hyponasty

v Why does human body become stiff after death ? Name that particular condition

vi How would you justify that amount of work a muscle does is reflected in changes in the muscle itself ?

vii How are identical twins formed ?

viii Differentiate between long day and short day plants

ix What are natural grass lands in the world are used for and how human activities are deteriorating these biomes ?

x What are taiga ? What kinds of environmental conditions are found there ?

xi Define the term Demography

xii Differentiate between reforestation and afforestation

Q3. (Marks 16) Write short answers any Eight (8) of the following

i Give names of two synthetic auxins with their effects

ii What are mechanoreceptors ? Give their role

iii What is role of limbic system in brain ?

iv Does jumping genes act as source of mutation ?

v Suggest how type A and AB parents could produce a child with blood group O

vi Define over dominance with an example

vii Define palindromic sequences

viii What is PCR ? Give its function

ix What is anther culture technique ? Give its role

x What is biome ? Give names of two biomes

xi Give importance of producers in ecosystem

xii How food chain is different from food web ?

Q4. (Marks 12) Write short answers any Six (6) of the following

i What is intercalary meristem ? Describe its role

ii Enlist some symptoms of aging

iii Describe the importance of promoter region during transcription

iv What are chromosomal aberrations ?

v Describe chemical composition of chromosome

vi Compare apoptosis with necrosis

vii How a cancerous cell differs from a normal cell ?

viii Differentiate between endangered and extinct species with examples

ix What is membrane invagination hypothesis ?

> Note: SECTION-II — Note: Attempt any Three questions from this section 8 x 3 = 24

Q5.

(A) What is kidney stone ? Describe its cure

(B) Define interphase , Explain its various sub-phases

Q6.

(A) What is bone fracture ? Describe the mechanism of their repair

(B) Define biogeochemical cycle. Discuss nitrogen cycle with labeled diagram

Q7.

(A) Write a detailed note on secretions of Adrenal glands

(B) What are endangered species ? What measures could be adopted for their preservation

Q8.

(A) Explain the human male reproductive system in detail

(B) Discuss genetic basis of ABO system in humans. Also give their importance in blood transfusion

Q9.

(A) Explain the role of nucleus in development with reference to Acetabularia

(B) What is genomic library ? How would you locate a gene of interest in the library

GROUP – II

OBJECTIVE

Time Allowed: 20 MINUTES | **Maximum Marks:** 17 | **Paper Code:** 8468

Note: You have four choices for each objective type question as A , B , C and D . The choice which you think is correct , fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. Commonly used restriction enzyme is

(A) PBR 322
(B) PSC 101
(C) Plasmid
(D) ECoR1

Q2. Eukaryotes are thought to have first appeared about

(A) 3.5 Billions
(B) 1.5 Billions
(C) 2.5 Billions
(D) 4.5 Billions

Q3. The change in frequency of allele at locus that occur by chance is

(A) Gene pool
(B) Genome
(C) Migration
(D) Genetic drift

Q4. Pick the biotic component from the following

(A) Animals
(B) Soil
(C) Water
(D) Atmosphere

Q5. Stone monuments are being eroded due to stone cancer by

(A) Green House effect
(B) Ozone depletion
(C) Acid rain
(D) Global warming

Q6. Incidence of uric acid kidney stone is

(A) 5 %
(B) 10 %
(C) 15 %
(D) 70 %

Q7. Which is stimulus for thigmotropism

(A) Touch
(B) Light
(C) Water
(D) Chemical

Q8. Clavicle connects scapula with

(A) Skull
(B) Femur
(C) Tibia
(D) Sternum

Q9. Hormone which promotes bolting of some rosset plants is known as

(A) Ethene
(B) Auxin
(C) Cytokinin
(D) Gibberellin

Q10. The 2nd largest part of brain is

(A) Thallamus
(B) Hypothalamus
(C) Cerebellum
(D) Cerebrum

Q11. In honey bee , males are haploid and produce sperms by

(A) Mitosis
(B) Meiosis
(C) Binary fission
(D) Multiple fission

Q12. Cleavage results in the formation of rounded closely packed mass blastomeres

(A) Gastrula
(B) Blastula
(C) Morulla
(D) Neurula

Q13. How many different kinds of t.RNA in human cell

(A) 54
(B) 45
(C) 25
(D) 20

Q14. The sequence of nucleotide that determine the amino acid sequence of a protein is

(A) Gene
(B) Allele
(C) Multiple allele
(D) Chromosome

Q15. Full cell cycle in yeast cell has length

(A) 30 minutes
(B) 60 minutes
(C) 90 minutes
(D) 120 minutes

Q16. A pure breeding tall pea plant was crossed to short plant. What will be the frequency of short plants in F1

(A) 0.25
(B) 0
(C) 0.5
(D) 1

Q17. Antibody made by soyabeen can be used as treatment for

(A) Genital Herpes
(B) AIDS
(C) Hepatitis
(D) Herpes simplex

SUBJECTIVE SECTION-I / SECTION-II

Q2. (Marks 16) Write short answers any Eight (8) of the following

i Describe some adaptations made by plants living in extreme dry conditions

ii How kidney helps to conserve water when body is facing dehydration ?

iii What are heterotherms ? Give two examples

iv Why ecdysis is necessary for most insects ?

v Describe the role of Ca^{+2} and ATP in muscle contraction

vi How snakes move from one place to another without legs ?

vii Compare parthenocarpy with apomixes

viii What is oestrous cycle ? Is it also present in humans ?

ix What do you mean by " Taiga " ? Give its conditions

x What are the main factors that determine productivity of an ecosystem ?

xi How global worming may effect human life on earth ?

xii Differentiate between renewable and non-renewable resources

Q3. (Marks 16) Write short answers any Eight (8) of the following

i How are synthetic auxins applied in agriculture ?

ii How does sodium potassium pump work in transmission of nerve impulse ?

iii Why insight learning is considered highest form of learning ?

iv What do you know about nullogamete ?

v Why AB blood group is known as universal recipient ?

vi A man is 45 years old and bald. His wife also has pattern baldness. What is the risk that their son will lose his hair ?

vii How do we obtain gene of interest ?

viii What is gene pharming ?

ix What do you know about Taq polymerase ?

x Define commensalism. Give example

xi What do you know about Autecology ?

xii Define Food Chain. Give an example

Q4. (Marks 12) Write short answers any Six (6) of the following

i Narrate the characteristics of dividing cells in plants

ii Give the effects of temperature on growth of plants

iii Draw a structure showing phosphodiester linkage

iv A human chromosome has a bulk of information. How ?

v How euchromatin and heterochromatin are different ?

vi What is the role of Actin and myosin in cell division ?

vii Write the characteristics of cancer cells

viii Give the contribution of Lamarck in evolution

ix Define gene pool and fixed allele

> Note: SECTION-II — Note: Attempt any Three questions from this section 8 x 3 = 24

Q5.

(A) Describe the major homeostatic functions of the liver

(B) Define mitosis. Write its importance

Q6.

(A) Describe vertebral column and rib cage

(B) Explain Nitrogen cycle with the help of sketch

Q7.

(A) Write a note on structure and function of fore brain

(B) The fossil record and comparative embryology are strong evidence of evolution. Justify

Q8.

(A) What is incomplete dominance ? Explain with the example of 4 O'clock plant

(B) Elaborate various components of female reproductive system

Q9.

(A) What is regeneration ? Why it is more common in some animals and not in others ?

(B) What is gene therapy ? Discuss its importance with two examples